

Algebraic and Transcendental functions

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Algebra skills

1. Use the base change formula to find the numeric value of $\log_2(e^2)$ without using calculator.
2. For the functions $f(x) = \frac{5x}{x^2+2}$ and $g(x) = \frac{1}{x}$ find the indicates composition and reduce the expressions as much as possible,
 - $(g \circ f)(x)$
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Use logarithmic properties to condense the following expression $2\log_8(y) + \frac{1}{3}[\log_8(x) - 2\log_8(w)]$.

Function analysis

Find the inverse function of $f(x) = \frac{2x-4}{4x^2+2x-10}$. Reduce the expression as much as possible and proof your answer by evaluation the function composition $(f \circ f^{-1})(x) = x$.

Modelling with math

Exponential decay The half-life of Radium-266 is 1600 years. Suppose that there is a sample of 1 kg. Recalling that the mass decays exponentially, fulfill the following points.

1. Write the function that models the remaining mass after t years.
2. How much mass will remain after 4000 years?
3. How much time will it take for the sample to be a third of its original value.

Algebra Toolkit **Algebra properties**

$$a + b = b + a,$$

$$(a + b) + c = a + (b + c),$$

$$a(b + c) = ab + ac,$$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd},$$

$$\frac{-a}{b} = -\frac{a}{b} = \frac{a}{-b},$$

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{ad}{bc}$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}, \quad \sqrt[n]{a} = a^{1/n}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

$$a^2 - b^2 = (a - b)(a + b)$$

Laws of exponents

$$a^r \cdot a^s = a^{r+s}, \quad (a^r)^s = a^{rs}$$

$$\frac{a^r}{a^s} = a^{r-s}, \quad (ab)^r = a^r b^r$$

$$\left(\frac{a}{b}\right)^r = \frac{a^r}{b^r}, \quad b \neq 0$$

Laws of Logarithms

$$\log_a(AB) = \log_a(A) + \log_a(B)$$

$$\log_a\left(\frac{A}{B}\right) = \log_a(A) - \log_a(B)$$

$$\log_a(A^C) = C \log_a(A)$$

Inverse functions property

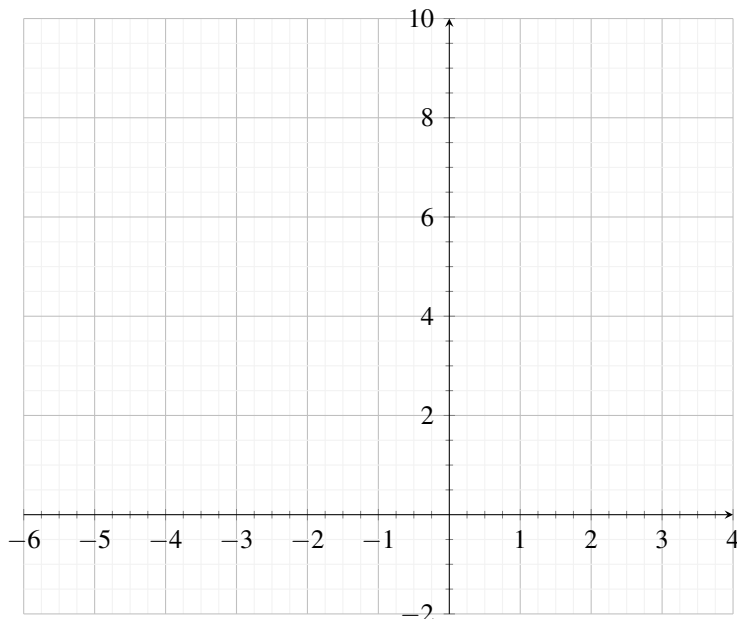
$$(f \circ f^{-1})(x) = x = (f^{-1} \circ f)(x)$$

$$\log_a(a^x) = x = a^{\log_a(x)}$$

Conic sections

General formula Identify the conic section give the following general equation $16x^2 - 9y^2 - 96x - 56y - 81 = 0$.

Graph Graph the following conic section $(x + 5) = 0.5(y - 5)^2$.

**Conic Sections**

Standard form of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Standard form of a parabola

$$(x - h) = 4a(y - k)^2, \text{ or } (y - k) = 4a(x - h)^2$$

Standard form of an ellipse

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

Standard form of a hyperbola

$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$$